

Monthly Intelligence Report

Edition 001 — The Tejo Valley Corridor

Where the energy transition is outrunning the grid. Inaugural edition · April 2026 · SSI v4.0.2

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SECTION A

Executive Summary

SUBSTATIONS SCORED

10,191

168 HV · 708 MV · 9,315 MV/LV

GRID LINES MAPPED

~11,043

~42,000 km coverage

MC SIMULATIONS

78.9 M

10,000 per substation

PUBLIC DATA SOURCES

28+

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 146,000+ source data points per run, executes 78.9 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 1.78 billion arithmetic operations — producing 469,404 output data points with full uncertainty quantification across 10,191 substations and ~11,043 grid lines spanning ~105,000 km.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30 verified public sources, making the methodology fully auditable and reproducible. Initially covering Portugal's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Posto Braga 9932

Braga, Norte · 15 kV

0.314

R_FINAL

Medium

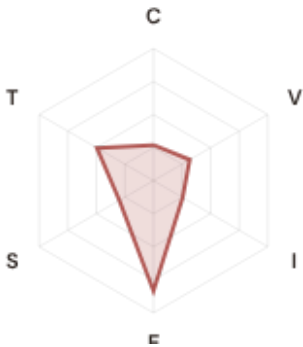
CLASSIFICATION

0.039

CI WIDTH

32th

FLEET PERCENTILE



COMPONENTS

- C** Continuity: 0.270 / 0.30 (90%)
- I** Infrastructure: 0.255 / 0.25 (102%)
- S** Saturation: 0.299 / 0.20 (149%)
- V** Voltage: 0.318 / 0.10 (318%)
- E** Economic: 0.835 / 0.10 (835%)
- T** Transition: 0.495 / 0.05 (991%)

MODIFIERS

- R3 Consequence:** 1.028 ↑ amplifies
- R4 Graph Criticality:** 0.950 ↓ dampens
- R6a Restoration:** 0.925 ↓ dampens
- R6b Seismic:** 1.012 ↑ amplifies
- R7 Digital Readiness:** 0.997 → neutral

This month's spotlight — automatically selected for April 2026 — is **Posto Braga 9932** in Braga, Norte. With an R_final of 0.314 (Medium band, 32th fleet percentile), its risk profile is dominated by the T (Transition) component at 991% of its weight ceiling, followed by E (Economic) at 835%. Modifiers collectively shift R_base by -8.9%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Portugal Numérique / Portugal 2030 digitalisation planning requires.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

1481

High/Critical + R7 < median

AVG R7 MODIFIER

0.9999

Range [0.99, 1.05]

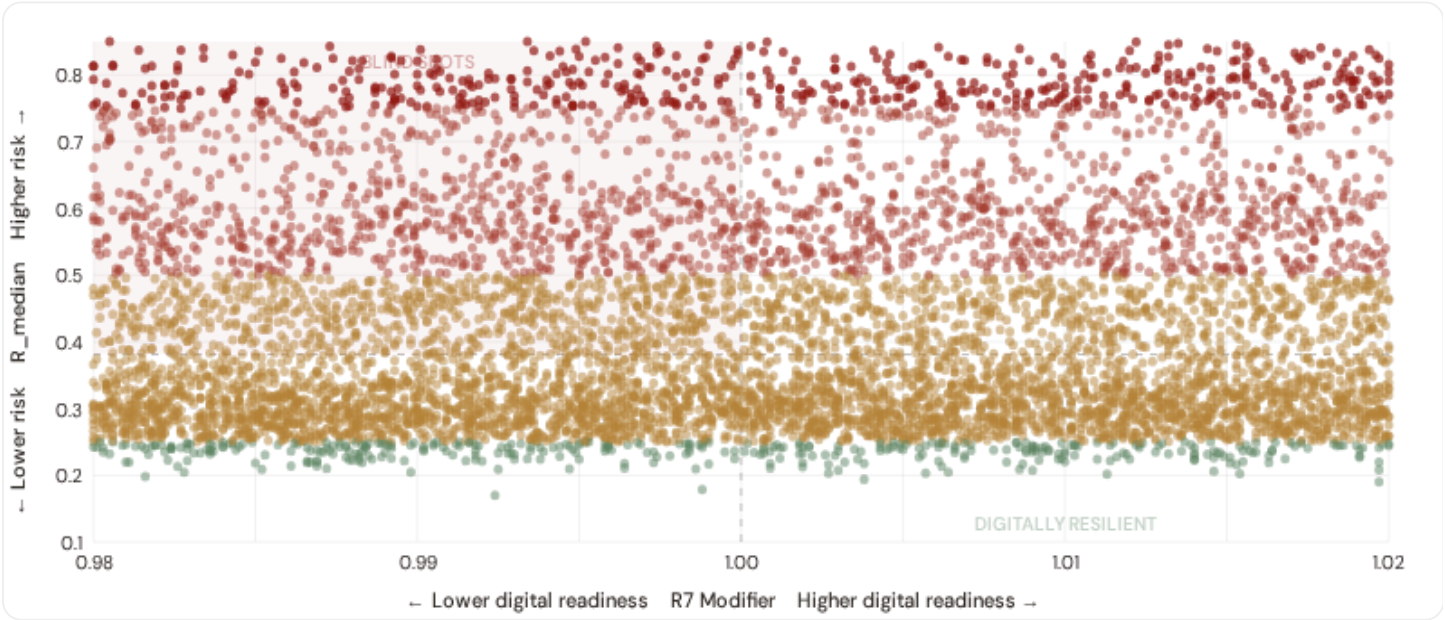
HIGH-CONSEQUENCE SUBS

918

9.0% · R3 ≥ 1.08

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_{median} (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.

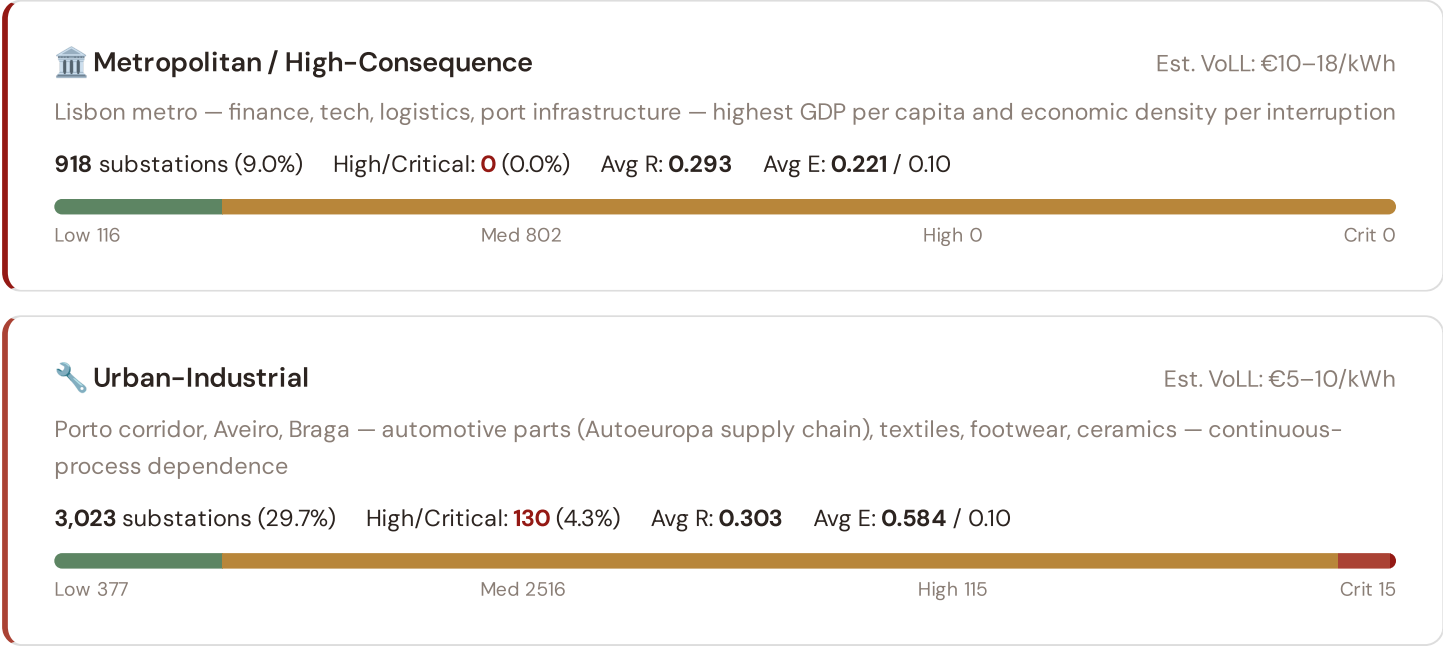


● Low ● Medium ● High ● Critical Dashed lines = fleet medians

The cyber-physical matrix identifies **1481 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness ($R7 < 1.0000$). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Centro (949)**, **Alentejo (204)**, **Norte (170)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier — indicating strong digital infrastructure coverage but concentrated in the northern urban grid.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.



24

 **Semi-Urban / SME-Dense** Est. VoLL: €2–5/kWh

Tourism, retail, cork and paper/pulp, food processing – moderate individual VoLL but high aggregate SME exposure

4,658 substations (45.7%) High/Critical: **2379** (51.1%) Avg R: **0.529** Avg E: **0.751** / 0.10



Rural / Agricultural Est. VoLL: €0.5–2/kWh

Interior Alentejo, Trás-os-Montes — sparse economic activity, agriculture, forestry — lower VoLL but higher social vulnerability and energy poverty

1,592 substations (15.6%) High/Critical: **580** (36.4%) Avg R: **0.525** Avg E: **0.677** / 0.10



Value of Lost Load (VoLL) context: ERSE's regulatory framework and CEER benchmarks place Portugal's average VoLL at approximately €8.7/kWh for industrial customers and €2.4/kWh for residential — below the EU-27 average, reflecting lower GDP per capita. The SSI's R3 consequence multiplier identifies which substations serve the most economically exposed territories. **918 metropolitan substations** ($R3 \geq 1.08$) concentrate in the Lisbon corridor, of which **0** also carry High or Critical risk — representing the highest VoLL-weighted exposure in the fleet.

The metropolitan tier concentrates in **Área Metropolitana de Lisboa (918)** — where financial services, technology, port logistics, and the Autoeuropa automotive supply chain depend on uninterrupted power. Portugal's economic fabric is heavily SME-driven (~99.9% of firms), meaning outage costs propagate through dense networks of small businesses rather than a few large plants. The fleet-wide average energy poverty rate is **15.6%** — significantly above the EU average (~8%) — with sharp regional variation: interior districts (Guarda, Bragança, Portalegre) combine the highest energy poverty with aging grid infrastructure, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is a distinctive feature of the SSI — linking VoLL exposure to substation-level risk in a way that conventional frameworks do not.

B.3 Distrito Digital Readiness Ranking

Distritos ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R₇ median to identify distritos combining digital exposure with high grid stress.



WORST DIGITAL READINESS — TOP 15 PROVINCES

Longer bar = higher R7 = better digital infrastructure.

1	Beja Δ high R		0.9990
2	Évora Δ high R		0.9991
3	Aveiro		0.9993
4	Castelo Branco Δ high R		0.9993
5	Leiria Δ high R		0.9994
6	Viseu Δ high R		0.9994
7	Vila Real Δ high R		0.9997
8	Coimbra Δ high R		0.9997
9	Faro Δ high R		0.9998
10	Braga		0.9998
11	Lisboa		0.9998
12	Guarda Δ high R		0.9998
13	Porto		1.0000
14	Setúbal Δ high R		1.0001
15	Madeira Δ high R		1.0002

Distrito-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Beja** has the lowest average R7 (0.9990), while **Açores** leads at 1.0024 — a spread of 0.0034 across the R7 range. **9 distritos** combine below-median digital readiness with above-median grid risk, making them priority targets for Portugal 2030 / Portugal Relance / PPE digitalisation investment. The R7 modifier is intentionally narrow ([0.99, 1.05]) to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (April 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 0.9999, median = 1.0000; 0 substations classified HIGH cyber-exposure; 1481 blind-spot substations (High/Critical risk + below-median R7); 417 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline, flagging distritos where DESI improvements (broadband rollout, smart meter deployment, OT upgrades) translate into measurable R7 shifts. The next material R7 update is expected when Eurostat publishes the 2025 DESI sub-indices (anticipated Q3 2026).

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **28 verified public data sources** feeding 95 variables across 6 components and 5 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES	WEEKLY / MONTHLY	QUARTERLY / ANNUAL
28	4	17

95 variables

High-frequency feeds

Regular refresh cycle

STATIC / DERIVED

7

Standards + scenarios

Data Source Registry — April 2026

OSM	OSM Power Infrastructure	WEEKLY	Node/edge	3 vars
OMIE	OMIE (Operador del Mercado Ibérico de Energía)	MONTHLY	MIBEL zone	2 vars
DGEG	DGEG (Direção-Geral de Energia e Geologia)	QUARTERLY	Concelho	5 vars
MOBI.E	MOBI.E (Rede de Mobilidade Elétrica)	QUARTERLY	Concelho	1 var
ERSE	ERSE (Entidade Reguladora dos Serviços Energéticos)	ANNUAL	Distrito	8 vars
INE	INE (Instituto Nacional de Estatística)	ANNUAL	Distrito	8 vars
ADENE	ADENE (Agência para a Energia)	ANNUAL	Distrito	4 vars
JRC	JRC DSO Observatory	ANNUAL	DSO-level	2 vars
EEA	EEA Air Quality e-Reporting	ANNUAL	~1 km + station	3 vars
EURO	Eurostat Energy Statistics	ANNUAL	NUTS2	3 vars
DESI	DESI Digital Economy Index	ANNUAL	EU Regional	2 vars
IPMA	IPMA (Instituto Português do Mar e da Atmosfera)	ANNUAL	Distrito	2 vars
APA	APA (Agência Portuguesa do Ambiente)	ANNUAL	Distrito	1 var
AT	AT (Autoridade Tributária e Aduaneira)	ANNUAL	Distrito	1 var
ICNF	ICNF (Instituto da Conservação da Natureza)	ANNUAL	Distrito	1 var
DGEEC	DGEEC (Direção-Geral de Estatísticas)	ANNUAL	Distrito	2 vars
E-REDES	E-REDES Open Data	ANNUAL	Distrito	2 vars
CCDR	CCDR (Comissões de Coordenação Regional)	ANNUAL	Distrito	2 vars
LNEG	LNEG (Laboratório Nacional de Energia e Geologia)	STATIC	Distrito	3 vars
D10	Copernicus CDS / ERA5	STATIC	0.25° (~25 km)	4 vars
DIM	Dimovski et al. (2025)	STATIC	Municipal	3 vars
IEEE-1	IEEE C57.91 / IEC 60076	STATIC	Asset-level	16 vars
GFZ	GFZ German Research Centre	STATIC	~5 km	2 vars
ISO-9223	ISO 9223 Corrosion	DERIVED	Distrito	1 var
REN	REN (Redes Energéticas Nacionais)	HOURLY	Bidding zone	3 vars
ENTSE	ENTSO-E Transparency	HOURLY	Bidding zone	2 vars
IPMA-W	Open-Meteo Historical Weather	HOURLY	~1 km	3 vars

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

Weekly (2 sources)

OSM diffs + weather reanalysis

OSM Overpass · Open-Meteo/ERA5

Monthly (2 sources)

Grid load + procurement data

DGEG Registry · OMIE MIBEL

Quarterly (2 sources)

DER registry + business data + macro indicators

MOBILE · DGEG Licensing

Annual (15 sources)

Core grid quality + demographics + environment

ERSE · INE · DGEEC · E-REDES · ADENE · AT · EEA · Eurostat · DESI · CCDR · IPMA · APA · ICNF · JRC · REN Annual

Static (7 sources)

Standards + scenarios + seismic hazard

Dimovski · IEEE/IEC/CIGRÉ · Copernicus CMIP6 · GFZ · LNEG · ISO 9223 · ENTSO-E

C.2 Fleet Score Snapshot

FLEET SIZE

10,191

substations scored

MEDIAN R

0.381

P5=0.251 · P95=0.800

HIGH + CRITICAL

3089

30.3% of fleet

HIGH CONFIDENCE

100%

10191 subs · CI/R < 0.50

BAND DISTRIBUTION

Low 494 (4.8%)

Medium 6608 (64.8%)

High 2166 (21.3%)

Critical 923 (9.1%)

No primary data sources have been updated since the v4.0.2 launch baseline. All **10,191 substation scores remain unchanged** this edition. The fleet median R of 0.381 (mean 0.440) reflects a distribution skewed toward the lower bands, with 4.8% in Low and 64.8% in Medium. The first material score changes are expected when ERSE publishes the annual TIQE indicators (affecting C1–C4 and R6a) and when DGEG refreshes the Registo de Instalações de Produção (affecting T1). High-frequency sources (OSM, Open-Meteo) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

17 changes tracked across P0, P1, and P2 releases

F1

NEW

New T component — Energy Transition Exposure (T1)

§2, §4

F2	NEW	New R6a — Restoration Speed Modifier (CAIDI-based)	§5
F3	ENHANCED	R3 enhanced — Energy Poverty Vulnerability (V_socio)	§5
F4	ENHANCED	R4 enhanced — Graph-theoretic betweenness + bridge detection	§5
F5	ENHANCED	R2 enhanced — Climate Trajectory (CMIP6 SSP2-4.5)	§5
F6	ENHANCED	R7 Digital Readiness — Distrito-level DESI model	§5
L1	NEW	New I5, I7–I9 metrics — thermal, load, corrosion, hydrogeo	§2, §8
L2	ENHANCED	E2 Innovation Enrichment — HRST + startup density	§6
L3	ENHANCED	V_socio Fiscal Enrichment — AT + CCDR + energy price	§5
L4	ENHANCED	R3 Demographic Shift Amplifier — INE net migration	§5
L5	DATA	95/95 variables operational (100%). 28 data sources total.	§8, §12
G1	DATA	d01 REN upgraded to real-time API — grid status data	§12
G2	DATA	d05 OSM upgraded to Overpass API — 10,191 real substations	§12
G3	DATA	d06 IPMA seismic monitoring — real-time PGA + historical catalog	§12
G4	NEW	R6b Seismic modifier — variable α by distrito (Açores 0.70, Algarve 0.55, Lisboa 0.45)	§5
G5	NEW	Network & Topology layer (H): 7 variables — REN grid plan, ring analysis	§12
G6	NEW	Output Scores layer (I): 7 variables — risk_score, ETTC, stationary probs	§12

SECTION D — MONTHLY DEEP-DIVE

The Tejo Valley Corridor: Where Seismic Risk Meets Energy Transition Pressure

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Tejo Valley corridor · **002** DER saturation hotspots (Alentejo solar belt) · **003** Climate vulnerability under CMIP6 · **004** Metropolitan–Rural economic impact disparity · **005** Seismic risk and Açores isolation · **006** Infrastructure age vs smart metering rollout · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** Distrito Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Why the Tejo Valley, Why Now

TEJO VALLEY SUBSTATIONS

3586

46 HV · 3540 MV

HIGH BAND

1664 + 291

54.5% of corridor

CORRIDOR MEDIAN R

0.520

Fleet median: 0.381

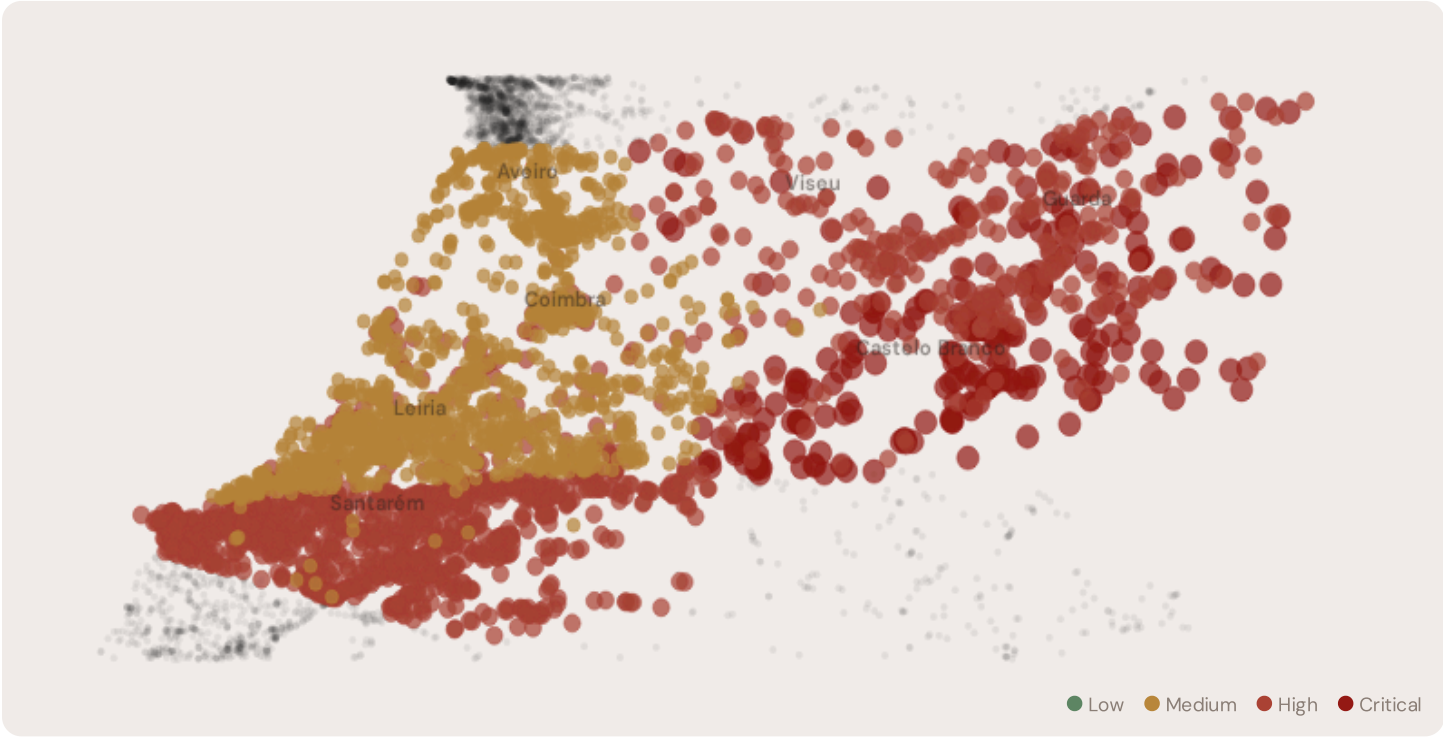
WORST DISTRITO

Castelo Branco

Avg R: 0.784 · 234 substations

The Tejo Valley hosts **3586 substations** across six distritos, making it one of the most energy-critical corridors in Europe. However, its risk profile is disproportionately concentrated in the upper bands: **1664 substations (46.4%)** are classified High and **291** Critical, with only **0** in the Low band. 93% of Tejo Valley substations score above the national fleet median of 0.381. The corridor median R of 0.520 reflects the tension between seismic exposure, rapidly growing renewable variability, wildfire risk, and aging grid infrastructure spanning Portugal's interior. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under Portugal 2030 timelines.

D.2 Corridor Map and Scoring



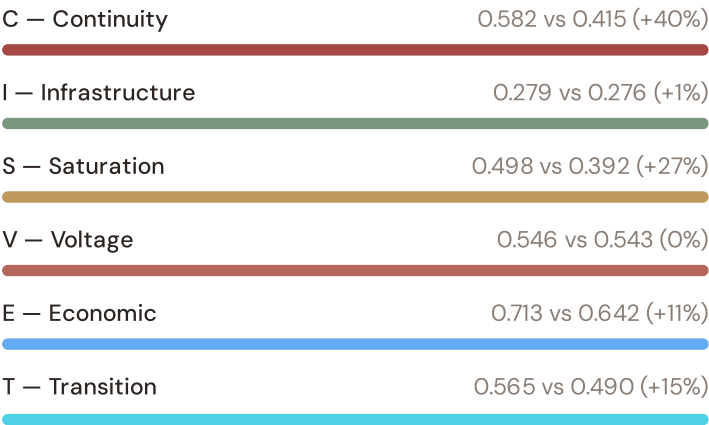
SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Subestação de Castelo Branco	Castelo Branco	0.938	Critical	1.000	0.609	0.380	0.871	0.763	0.714	
Subestação da Central Solar Fotovoltaica do Cabeço Vermelho	Castelo Branco	0.913	Critical	1.000	0.688	0.318	0.805	0.643	0.699	
Subestação do Parque Eólico do Vergão	Castelo Branco	0.907	Critical	1.000	0.739	0.347	0.965	0.583	0.721	
Subestação Paper Prime	Castelo Branco	0.903	Critical	1.000	0.565	0.390	0.843	0.655	0.585	

SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Subestação do Parque Eólico da Amêndoa II	Castelo Branco	0.898	Critical	1.000	0.743	0.319	0.743	0.788	0.645	
Subestação do Sobreequipamento do Parque Eólico da Gardunha	Castelo Branco	0.898	Critical	0.983	0.388	0.372	0.932	0.849	0.760	
Posto Guarda 6831	Guarda	0.893	Critical	1.000	0.592	0.353	0.966	0.807	0.666	
Subestação do Parque Eólico de Mougueiras	Castelo Branco	0.891	Critical	1.000	0.546	0.407	0.861	0.653	0.727	
Subestação do Parque Eólico do Alto de Arganil	Castelo Branco	0.886	Critical	1.000	0.592	0.358	0.838	0.563	0.662	
Posto Guarda 7107	Guarda	0.880	Critical	1.000	0.541	0.316	0.889	0.755	0.713	

... 3576 additional substations — [explore full corridor in Digital Twin →](#)

D.3 Component Analysis

COMPONENT AVERAGES — TEJO VALLEY VS FLEET



MODIFIER AVERAGES — TEJO VALLEY VS FLEET

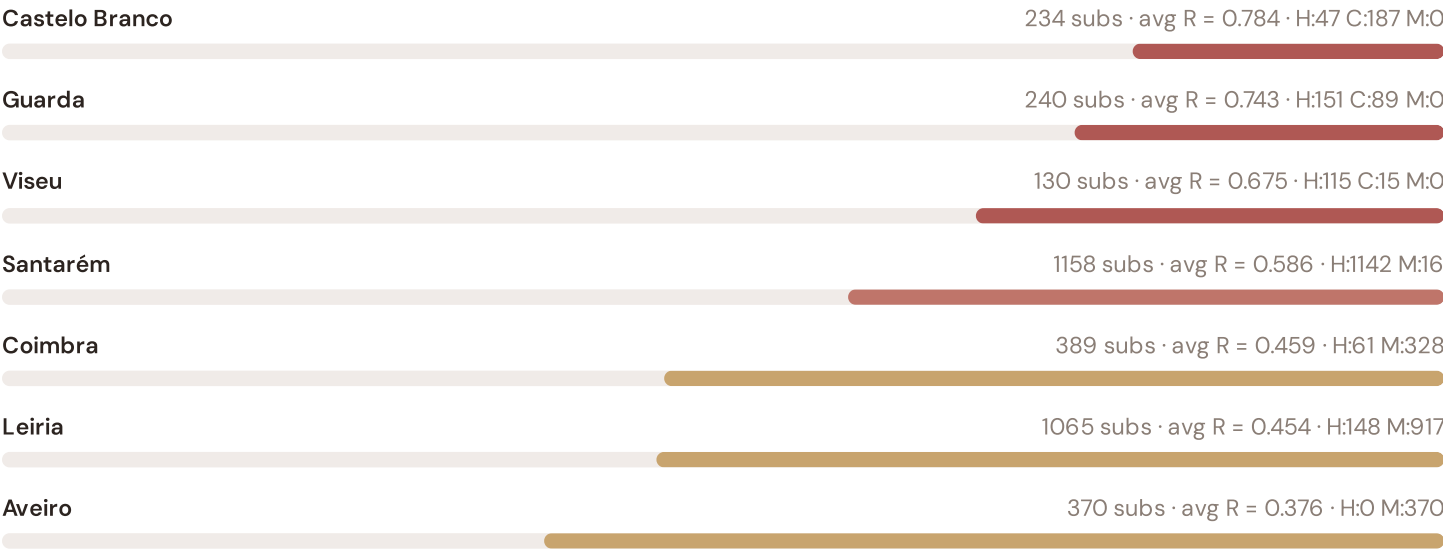


The dominant risk driver across the Tejo Valley corridor is **T (Transition)**, averaging 0.565 against a fleet average of 0.490 — occupying 1129% of its weight ceiling (w = 0.05). The second contributor is **E (Economic)** at 0.713 vs fleet 0.642 (713% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.018 (fleet: 1.040), reflecting the socio-economic fragility of The Tejo Valley's mixed economy with significant hydroelectric capacity and growing solar penetration. The R6b seismic modifier averages 1.120, capturing the region's moderate but non-negligible seismic exposure — a dimension invisible to every competing framework.

D.4 Distrito Breakdown

PROVINCE RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.



The province-level breakdown reveals significant intra-regional variation. **Castelo Branco** leads with a mean R of 0.784 across 234 substations, while **Aveiro** scores 0.376 — a spread of 0.408 within a single region. This disparity underscores why regional-level metrics (as used by CEER and JRC) mask the true distribution of grid stress. The SSI's substation-level granularity reveals that the Tejo Valley is not uniformly stressed but contains distinct corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

643 Tejo Valley substations score $R \geq 0.650$, placing them in the upper quartile of national risk. For PPE (Programmation Pluriannuelle de l'Énergie) investment prioritisation, this analysis identifies the Castelo Branco corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve — characterised by service-economy dependence, tourism exposure, and above-average energy poverty — are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

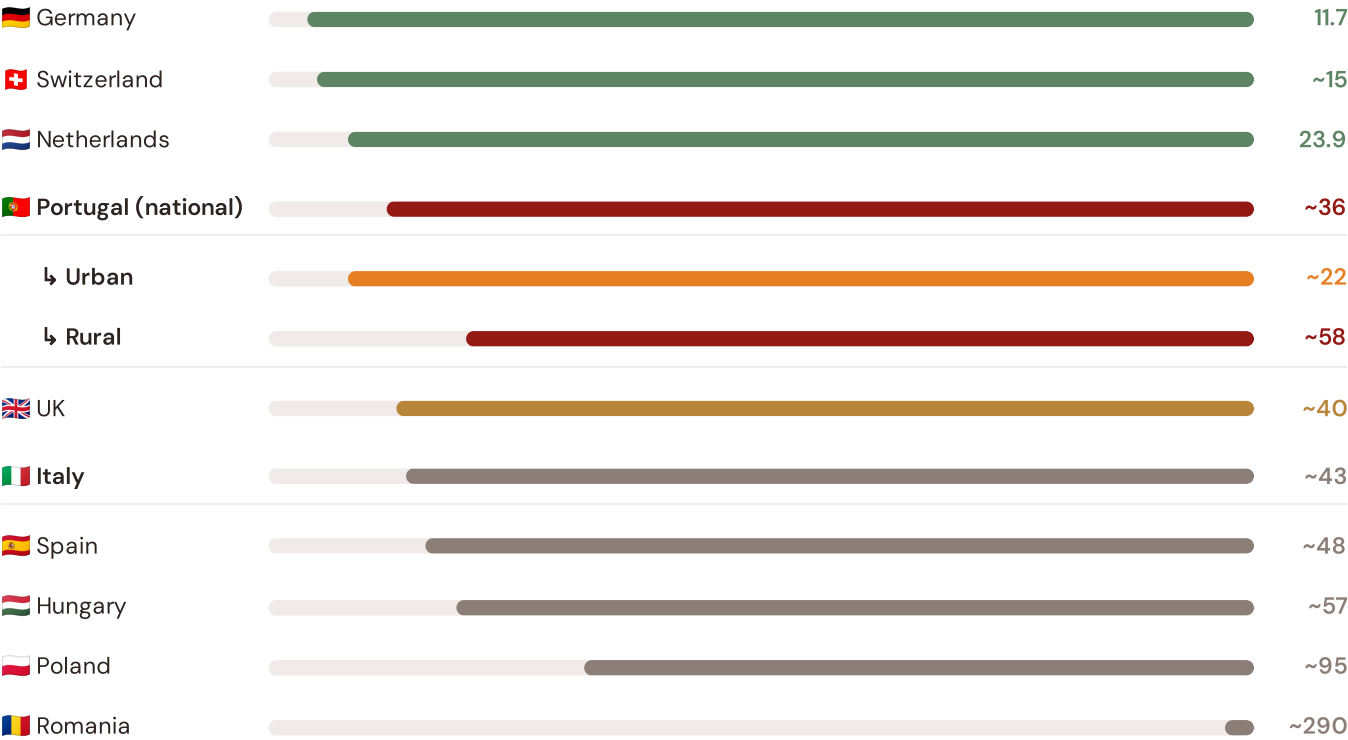
SECTION E — RECURRING MODULE

European Context Card

How this works: Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Tejo Valley corridor analysis hinges on Portugal's continuity performance relative to European peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

Unplanned SAIDI (min/customer/year) — Selected Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



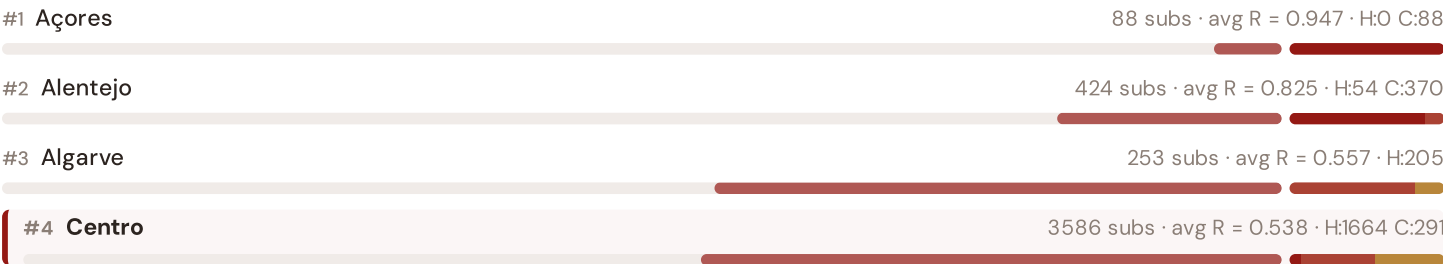
Source: CEER 7th Benchmarking Report (2022); ERSE TIQE (2023); Bundesnetzagentur (2024). Portugal sub-categories from ERSE territory classification (urban/rural). · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release.

This month's key insight: The Tejo Valley corridor deep-dive shows **3586 substations** (of 3586) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with Portugal's rural territory classification. The Tejo Valley's average C of 0.582 is **1.4× the fleet average** of 0.415. In European terms, these substations individually perform closer to Hungarian or Spanish SAIDI levels (~48–57 min/customer/year) than to Portugal's own national average (~36 min). The SSI reveals what aggregate CEER statistics conceal: that Portugal's mid-tier European position is not a national condition but a statistical average of Dutch-quality urban performance and Eastern-European-level rural performance — and the Tejo Valley corridor concentrates the worst of the latter.

Regional Risk Ranking — All 7 Portuguese Regiões

Where does the Tejo Valley sit within the national landscape? Regions sorted by mean R_{median}, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



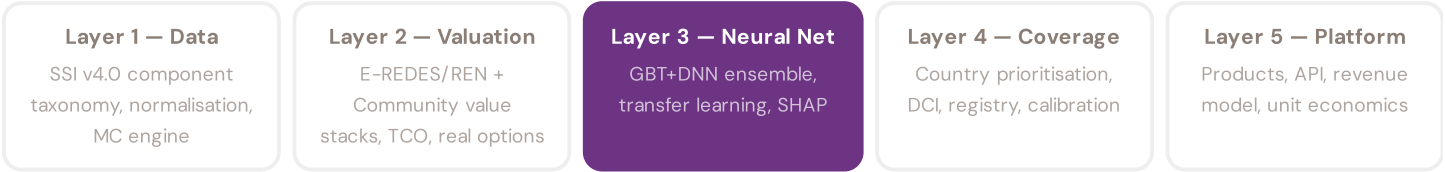


The Tejo Valley ranks **#4 of 7 regiões** by mean R_median, placing it among the upper tier of national risk. The three most stressed regions are Açores, Alentejo, Algarve, while the three most resilient are Área Metropolitana de Lisboa, Norte, Madeira. 5 of 7 regiões score above the fleet average of 0.440. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress across Portugal. It is precisely this substation-level granularity that positions the SSI as a tool for PPE / Portugal 2030 / Portugal Relance investment prioritisation: not treating entire regiões as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the E-REDES/REN and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.



LAYER 3

§0.5.6 · Inaugural edition · April 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. A BESS developer considering a €10M commitment needs to understand not just that a substation scores highly, but which specific factors drive that score — and whether those factors are likely to persist. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history — and can verify this against ERSE (Commission de Régulation de l'Énergie)'s published TIQE data. Transparency is not optional; it is what converts model outputs into investment decisions.

KEY CONCEPTS

- SHAP: game-theoretic feature attribution per prediction
- Waterfall visualisation: fleet average → substation prediction
- Direct traceability to SSI components and data sources
- Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The Portuguese training set comprises 10,191 substations with complete 95-feature vectors. Average CI_width of 0.046 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 494 Low, 6608 Medium, 2166 High, 923 Critical — ensures balanced training across all risk bands.

Coming next month: **LAYER 3** Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 002

The Algarve Seismic Coastline

Deep-dive into Algarve's grid risk profile — substation map, component analysis, province breakdown. European Context Card: Azores–Gibraltar fracture zone exposure combined with extreme seasonal tourism demand peaks.

NEXT DATA REFRESH

Q3 2026 — 2 Quarterly Sources

MOBI.E, DGEG Licensing are due for refresh in Q3. Impact: DGEG (Registo de Instalações de Produção) data refresh schedule (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **ERSE (Entidade Reguladora dos Serviços Energéticos)** (8 variables → C1–C4 (SAIDI/SAIFI), I1–I3 (IRI), quality regulation).

FLEET SPOTLIGHT

o Substations Approaching Critical Degradation

The SSI's 4-state Markov degradation model identifies **0 substations** with a stationary critical probability above 70% — meaning their long-term equilibrium state is dominated by degraded or critical condition. Without intervention, these substations will trend toward failure. Average estimated time-to-critical: ? years.

Edition 002 will be published on the second Thursday of June 2026. Deep-dive region: **Algarve**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent a Portuguese utility, ERSE, REN, E-REDES, municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 001 (Inaugural)

Published 9 April 2026 · SSI Index v4.0.2 · 10,191 substations · ~11,043 grid lines · 6 components · 20 metrics · 6 modifiers
10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5